

(DIS1 through DIS8). The port 1 and port 3 are used for scanning the keyboard buttons, except switch S6. Transistors (T1 through T8) and resistors (R1 through R8) are used for digit selection. The 7-segment

displays used are LTS542, which are common-anode type. Eight current-limiting resistors (R9 to R16) of 100Ω each are used.

The pin connections between the matrix keyboard and IC1, with key

functions, are shown in Table III.

The multiplexing of eight 7-segment displays is done by using unconventional queer multiplexing technique. The even and odd segments are connected differently in this technique

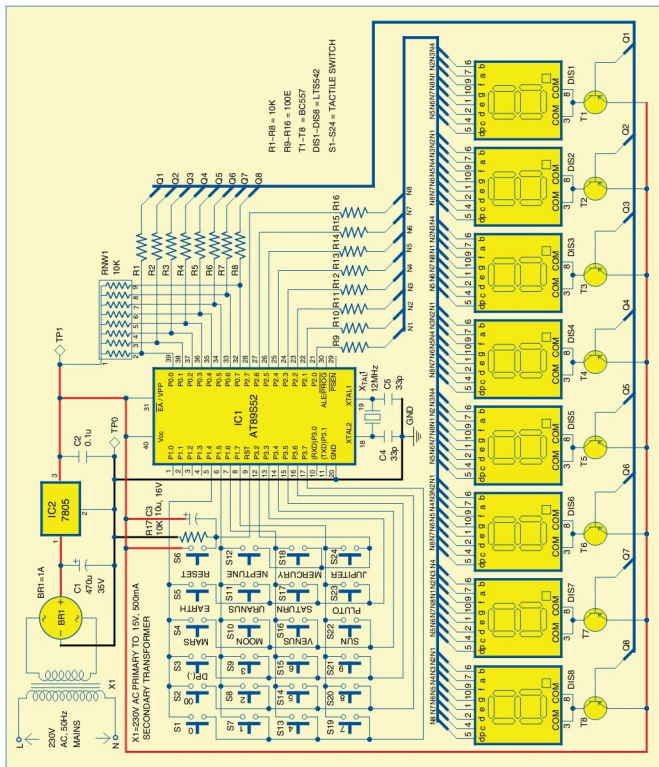


Fig. 2: Circuit diagram of the celestial weight calculator